



Software Design with Figma

BIS605

Software Development Technologies for Digital Business

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Resources

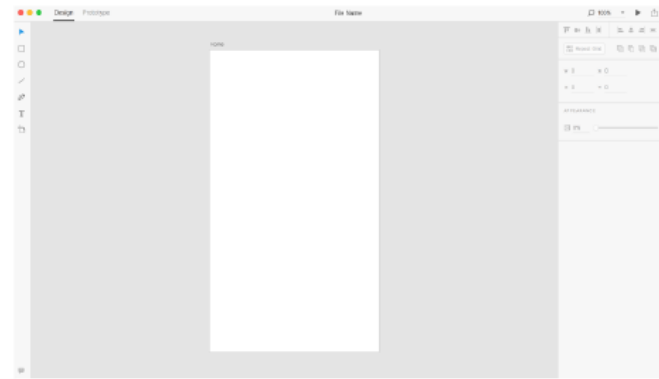
- <https://www.figma.com/>
- <https://www.csd.uoc.gr/~hy364/files/lectures/The%20Figma%20design%20tool.pdf>
- <https://applab.unc.edu/figma-tutorial.pdf>

Prototyping Tools

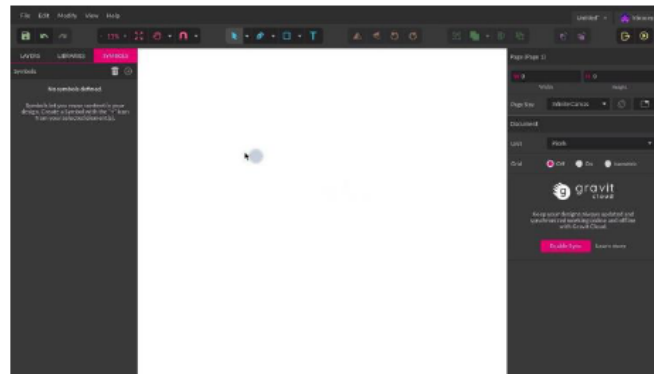
Figma



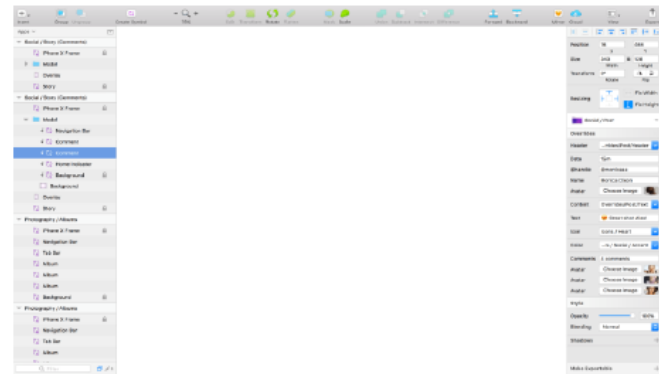
Adobe XD



Gravit Designer



Sketch



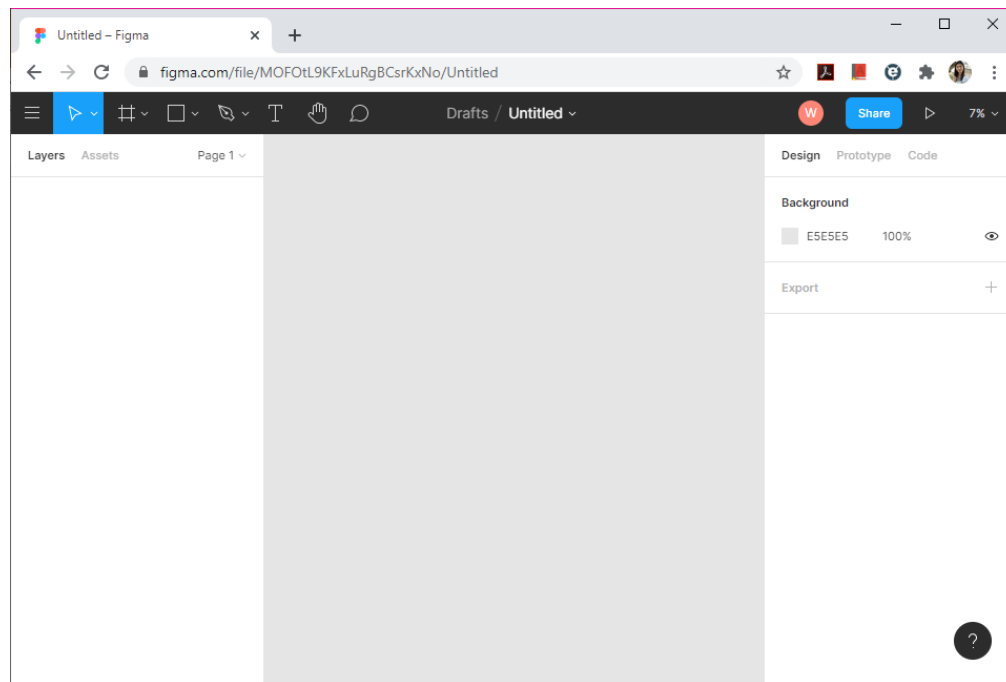
Figma

- Design tool which allows collaborative design
- Online and Desktop
- Fast and easy to use

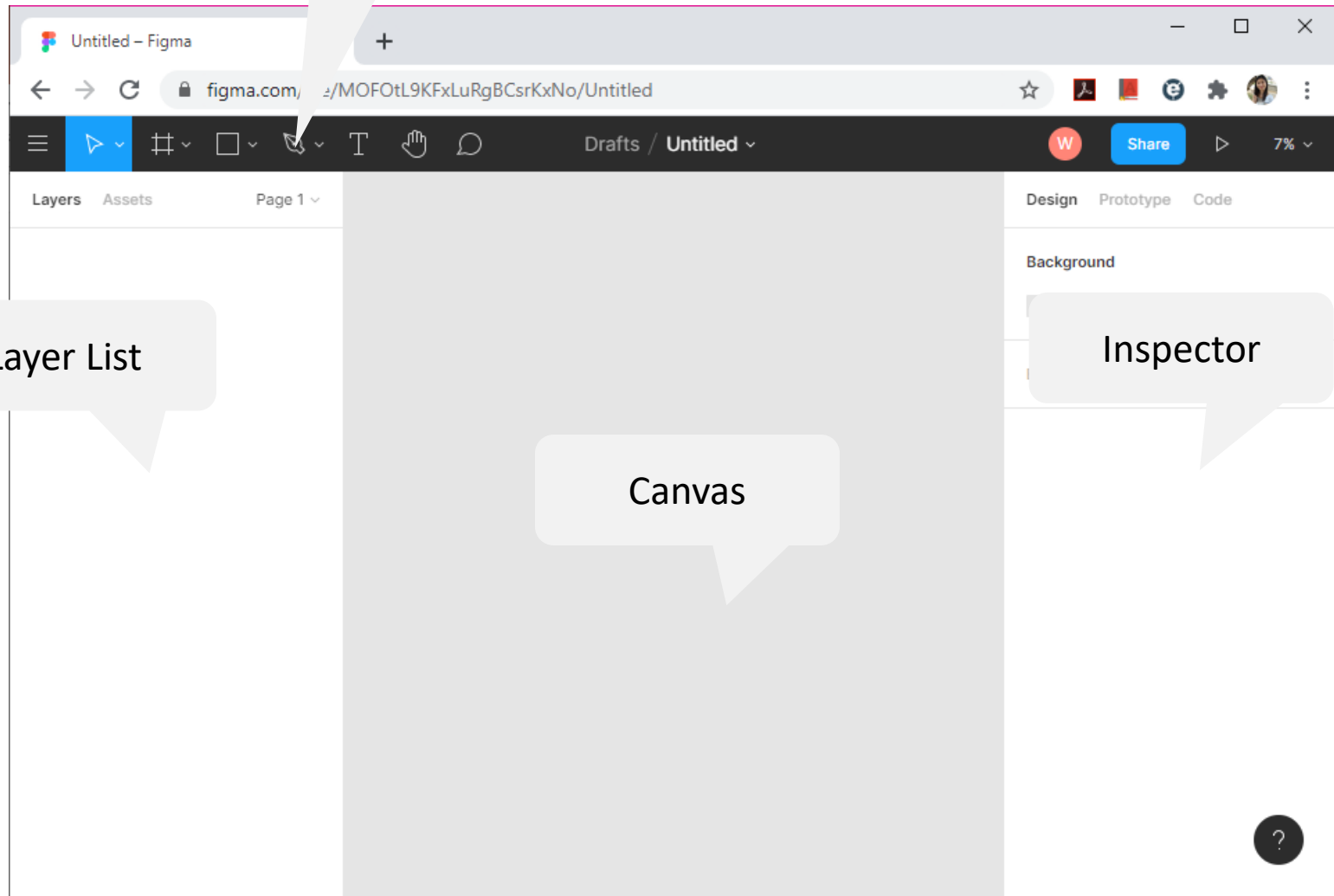


Setup

- Goto <https://www.figma.com> and create an account
- Create a new project using + icon and create a new File
- The new File will appear:

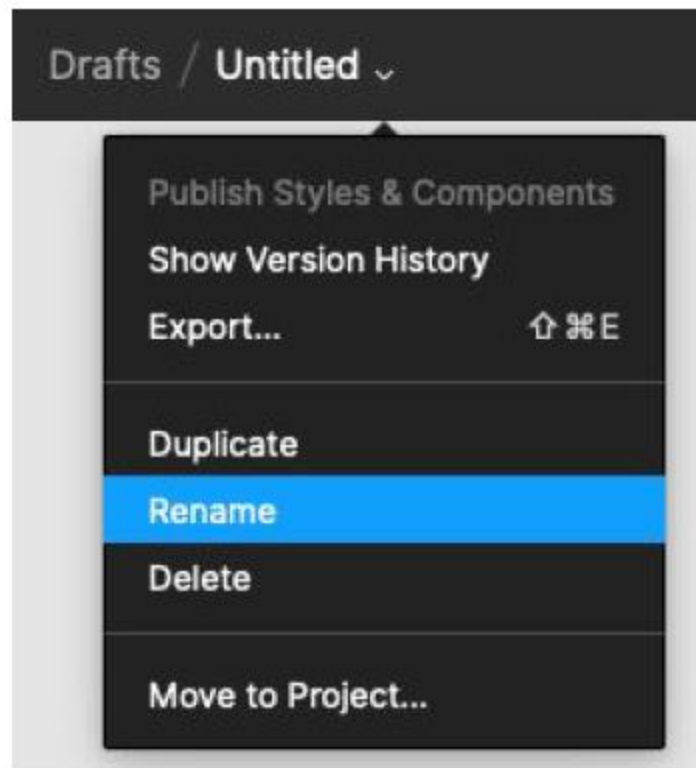


Setup



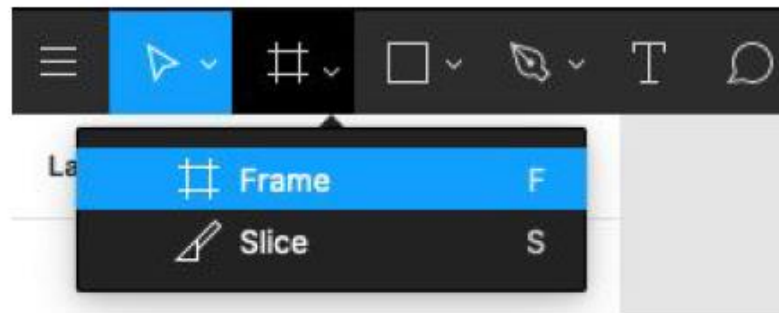
Setup

- Click the arrow in the top-middle of your screen to rename your project to “Calculator”

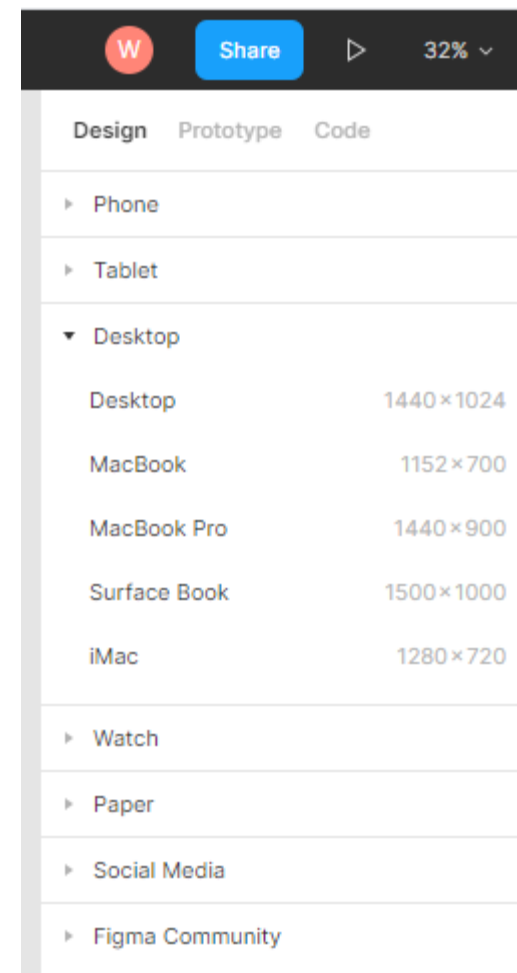


Frame

- Select Frame tool:



- You can create a frame by:
 - Dragging on the canvas
 - Use pre-define frame size appear in the inspector

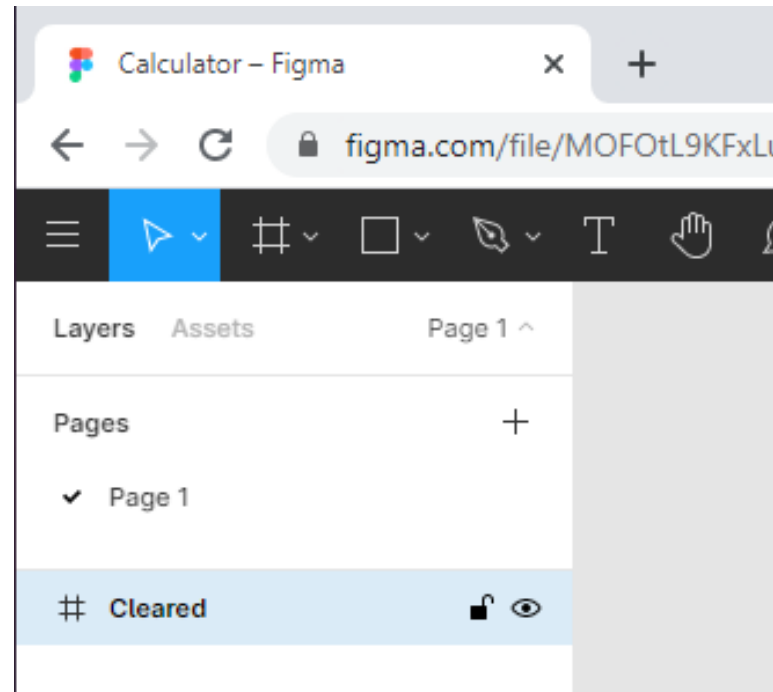


Inspector

- We will use Design tab of the inspector.
- You can use inspector to
 - resize the frame
 - Change color and other features
- Resize the frame to
 - Width: 283
 - Height: 324
- Change the background color to #B2B2B2
- Set the X and Y to 0

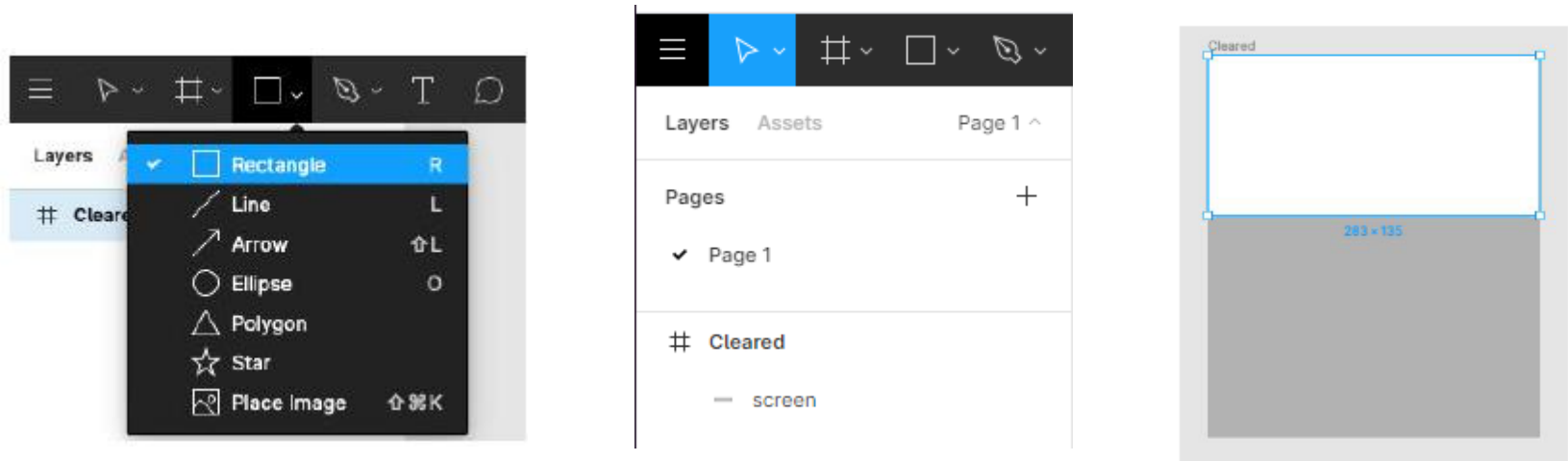
Layer

- Double Click on the frame that you created and rename it to “Cleared”. This is the name we will use for this screen (frame)



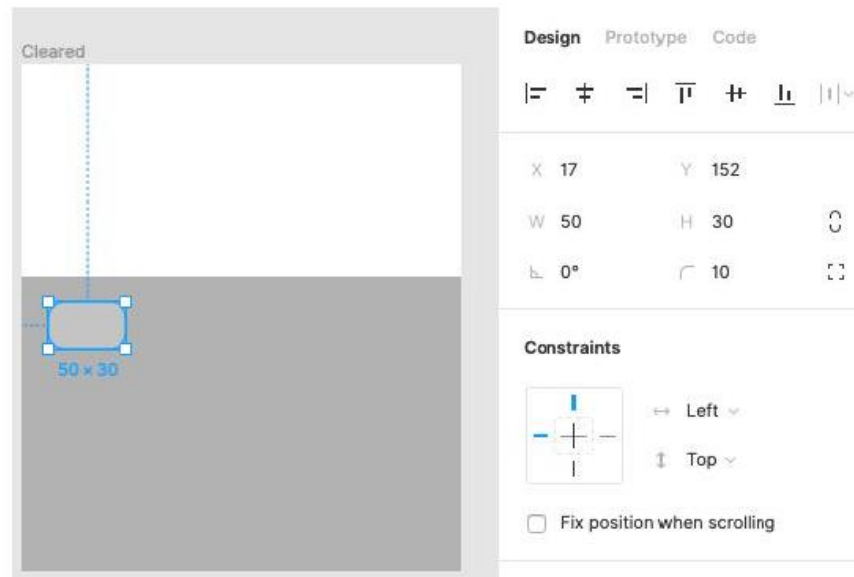
Adding the Calculator Display

- Select the rectangle tool from the toolbar.
- Drag an area on the canvas to add a rectangle. This will be the screen of the calculator to display numbers, operations, and results. Rename it as screen.
- Using the inspector, set the rectangle's X to 0, Y to 0, width to 283, height to 135, and color to #FFFFFF (i.e. white).



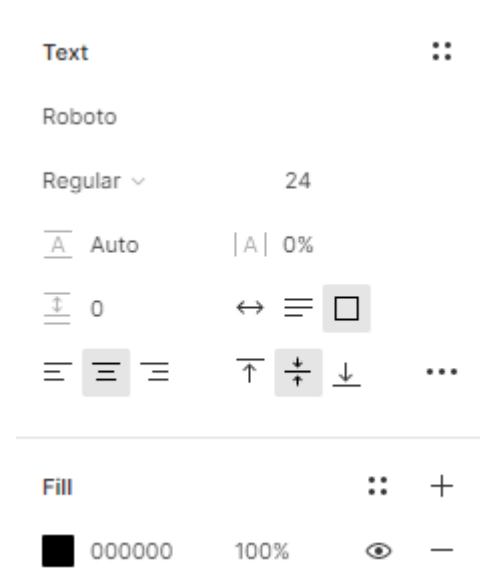
Add the First Button

- Add a smaller rectangle for the calculator buttons.
- Set the width to 50 and height to 30. Use your arrow keys to set X to 17 and Y to 152.
- Set the color to #C4C4C4. Set the border radius to 10.



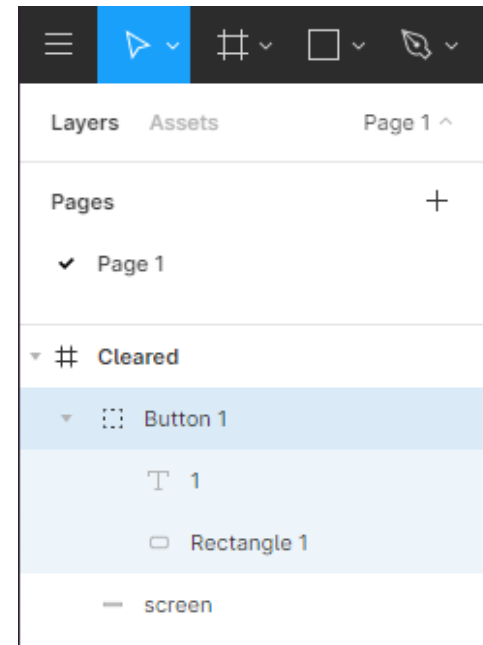
Add Label

- Select the text tool from the toolbar.
- Add a text label on the button you just created, then type “1”
- Using the inspector, set the label’s horizontal alignment to center, vertical alignment to middle, font to Roboto, and size to 24, as shown to the right.



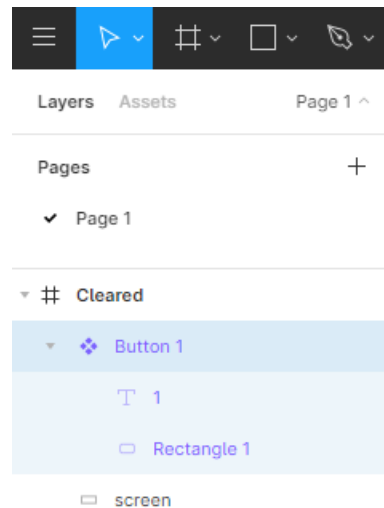
Grouping Elements

- Find the button and the label in the layer list.
- Select them both by holding control (Windows) or command (macOS) while clicking. Right click the selection and select “Group Selection”. Double click on the new group in the layer list to rename it to “Button 1”.



Make a Component

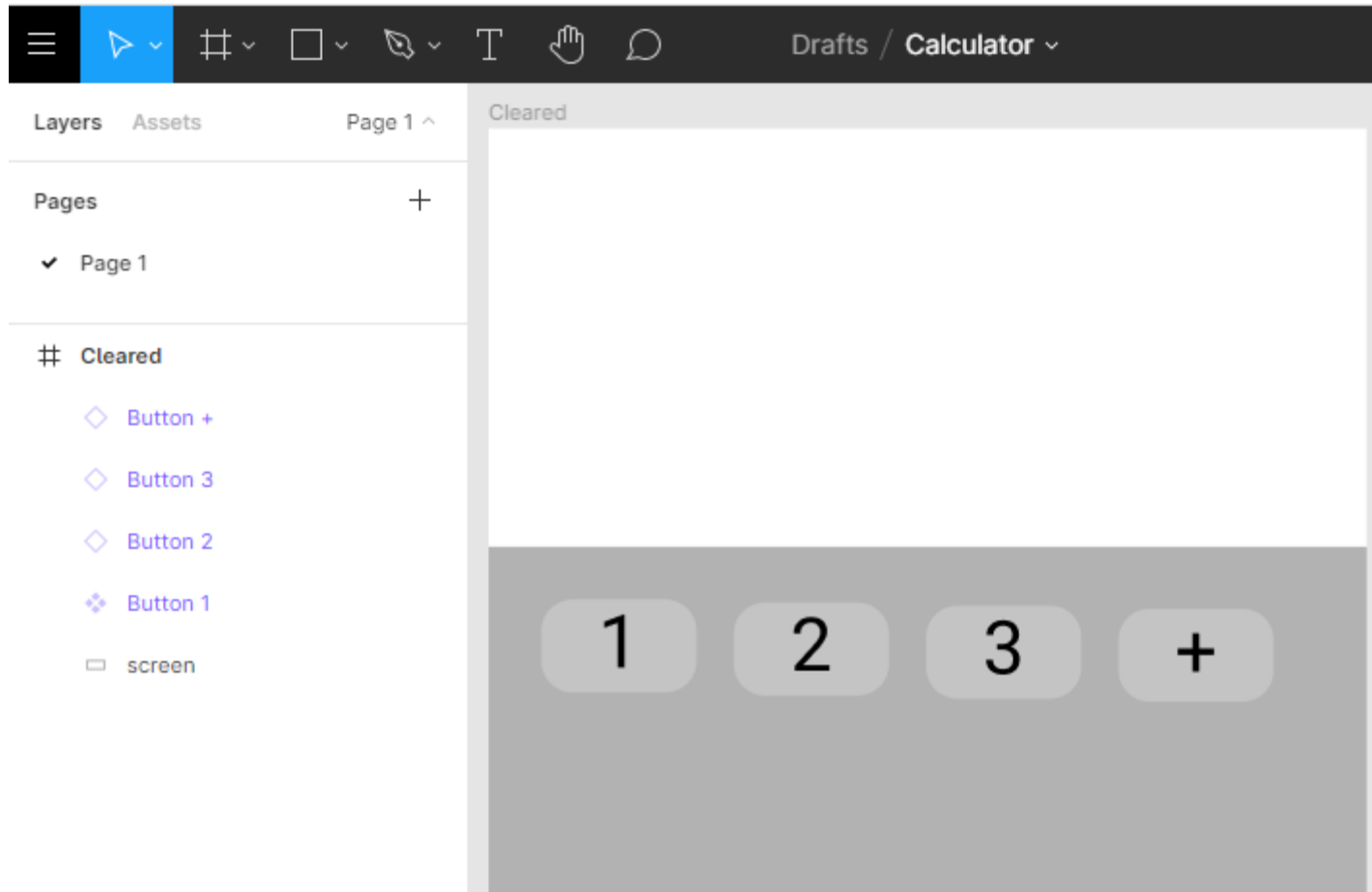
- Components are elements you can reuse across your designs. They help to create and manage consistent designs across projects.
- To create component, select group “Button 1”, then click “create component” button at the toolbar. 



Duplicating Elements

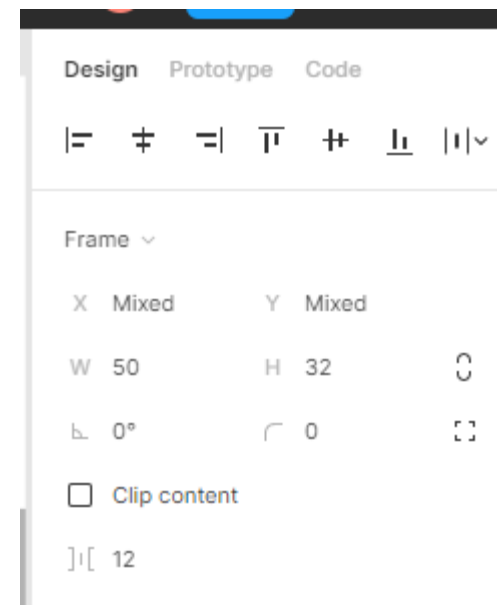
- Select the component “Button 1”, then press control-D (Windows) / command-D (macOS) to duplicate the group.
- Move the new group to the right by clicking and dragging the element in the canvas.
- Rename the group “button 2”, and change the text of the label to 2. Repeat to create a new button 3 and a new button +.

Duplicating Elements



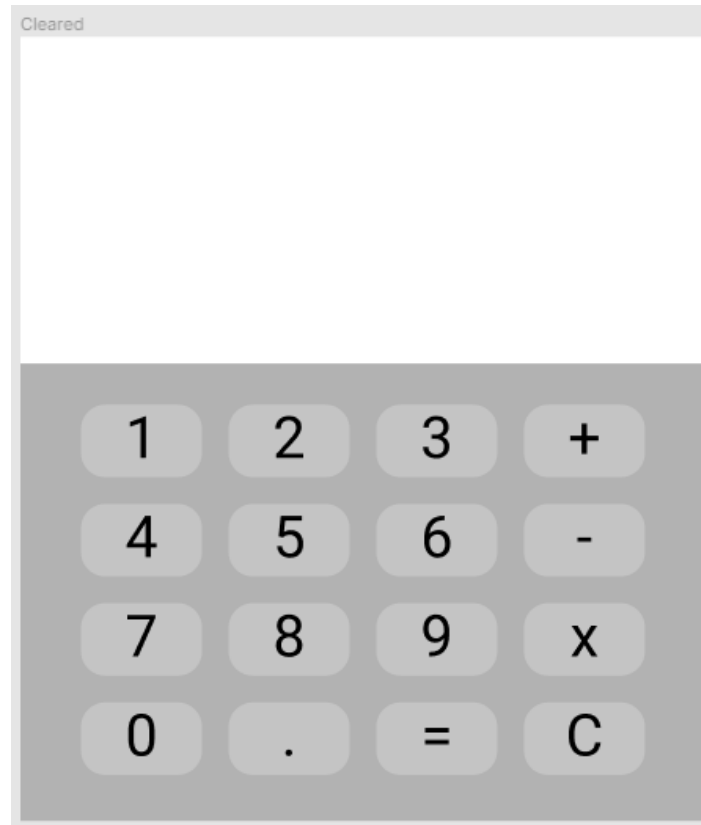
Positioning Elements

- Select all four buttons. (Drag mouse cover all buttons)
- At the alignment panel, there are many option for alignments.
- Click “Distribute Horizontal Spacing”



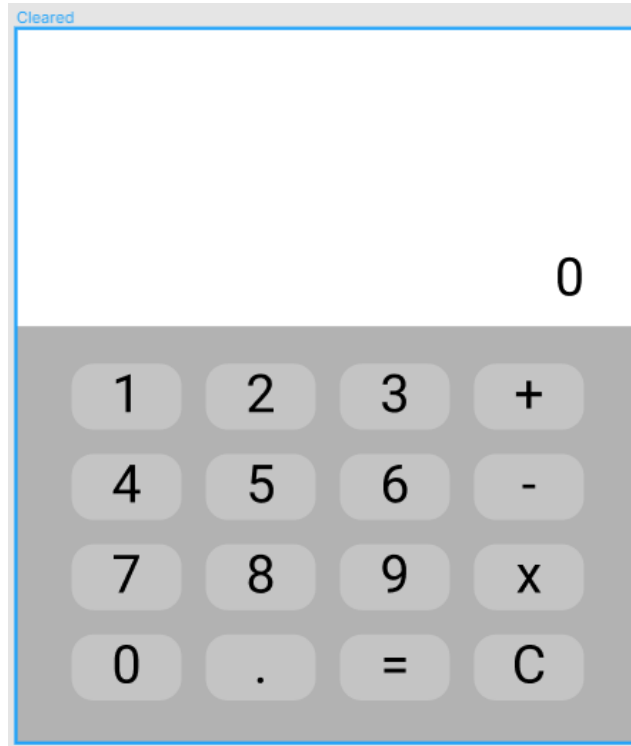
Duplicating Rows

- Apply your learning so far to fill out the remaining 3 row of buttons.



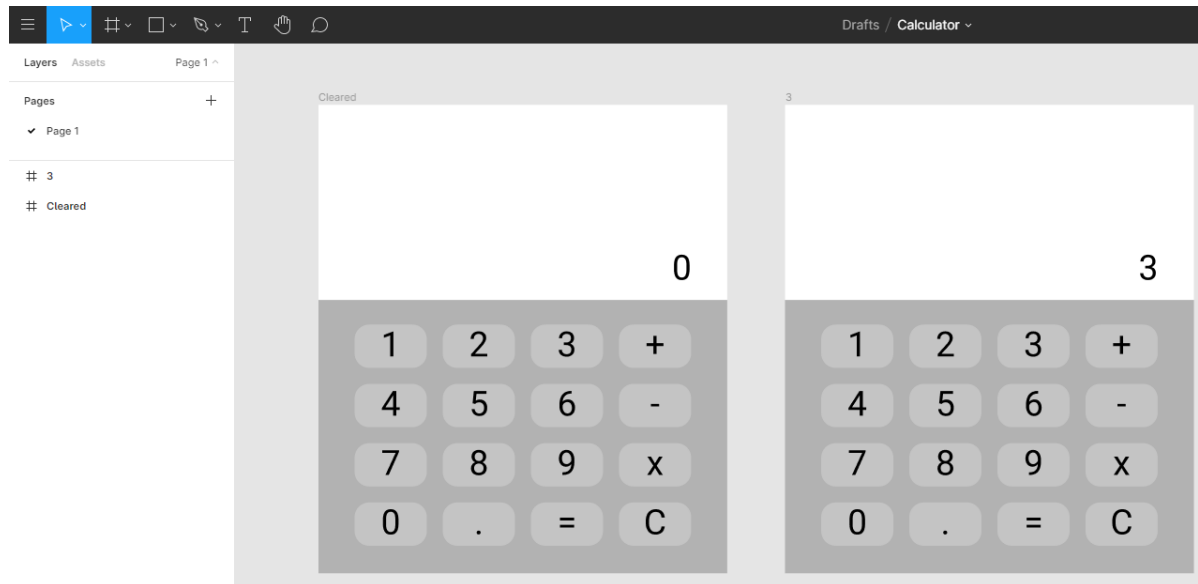
Add Display Label

- In the initial state, our calculator displays 0. Add a text element 0 to the display screen, sized the same as the rectangle.



Create Other Screens

- In the layer list, select the top-level frame, which you named “Cleared” in Step 3.
- Duplicate the frame, move it to the right of the first frame, change the display label to “3”, and rename the frame to “3”.



Create Other Screens

- Create other screens whose labels and displays are “3+”, “3+3”, and “6”, respectively, for a total of 5 screens.

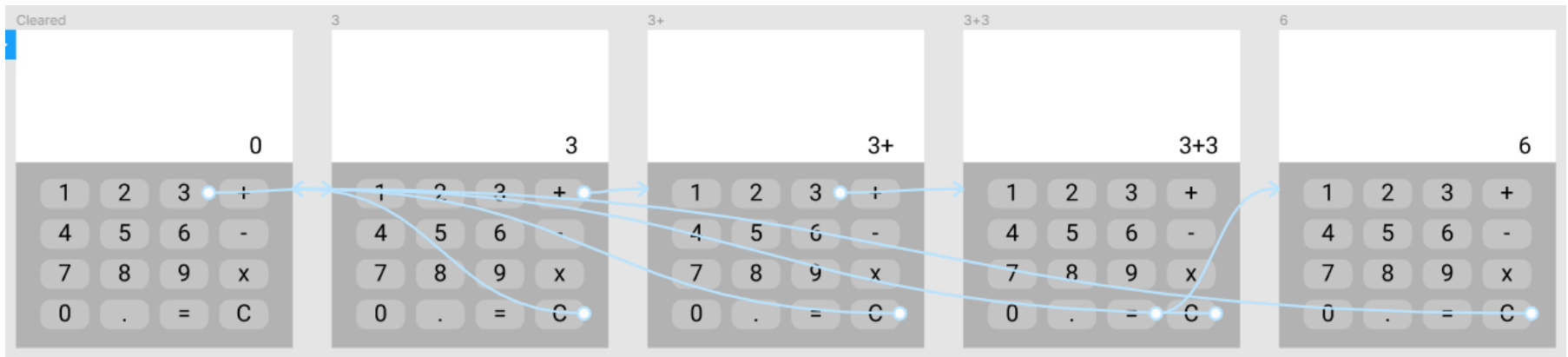


Create Linkable Prototype

- Select the first frame (named “Cleared”), and select the “prototype” tab of the inspector.
- Select Button 3 within the frame. Notice the blue circle on the right edge of the button. Drag that circle to the frame labeled “3”.
- Likewise, connect the “+” button in the “3” Frame to the “3+” Frame, and likewise for the next “3” button and the following “=” button. Finally, connect the “C” button of all but the first frame to the “Cleared” frame.

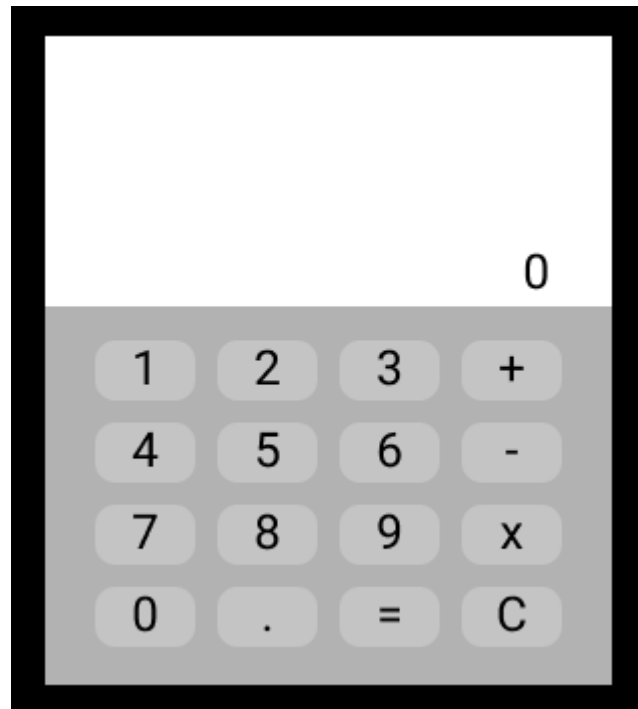
Create Linkable Prototype

- After deselecting all frames, your prototype connections should look like this:



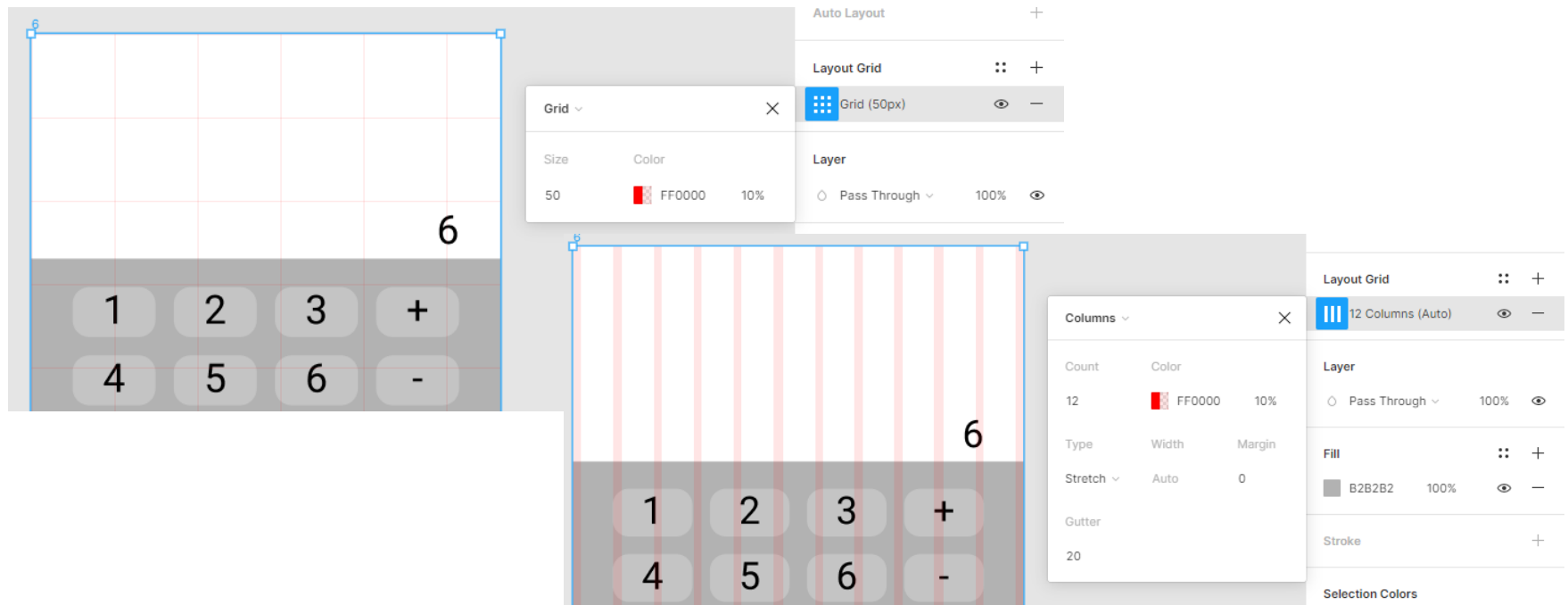
Present Prototype

- Click the “play” triangle icon in the upper right of the Figma UI to see the clickable prototype in action.



Grid

- You can use grid layout to help the alignment.
- In the inspector under the Layout Grid Section.
- You can define grid/column/row as you want.

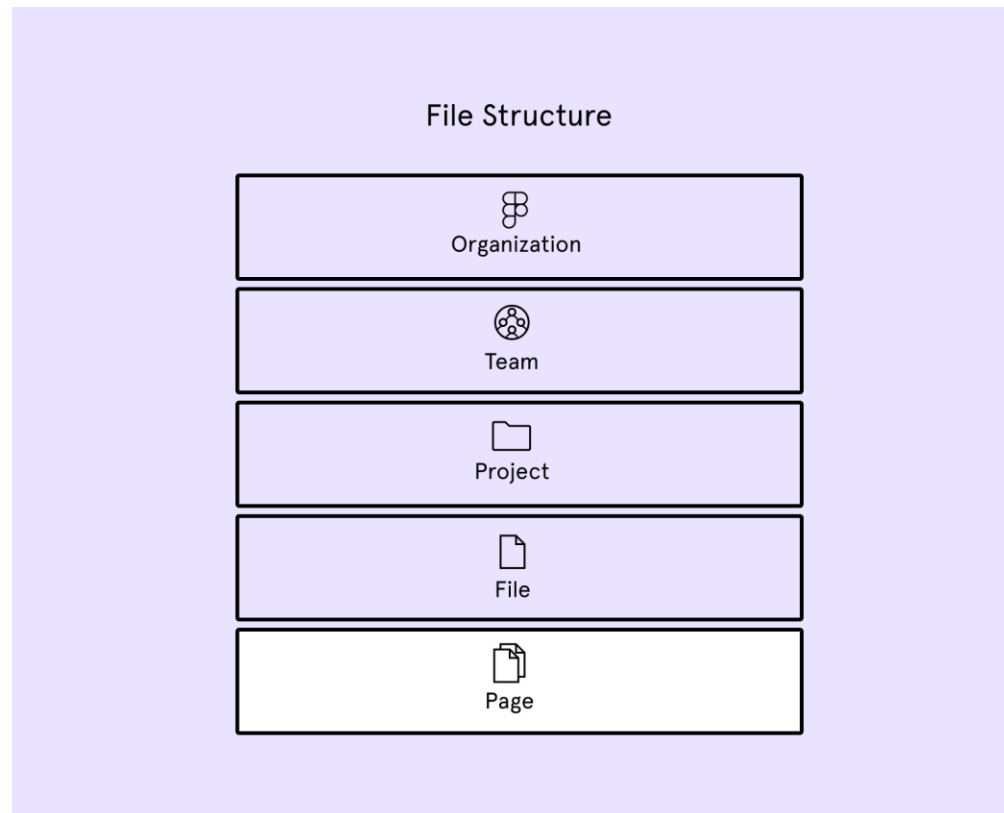


Add Image

- You can add image several ways.
 - Drag and drop images onto the Canvas
 - Import files via the File Browser (Select “File” → “Place Image”)
 - Add animated GIFs to prototypes
 - Add images to objects in bulk with place image
 - Upload an Image as a fill
 - Copy and paste images between objects
 - Paste images from your clipboard
- For more info:
 - <https://help.figma.com/hc/en-us/articles/360040028034-Add-Images-to-your-designs>

Page

- There are five levels of File structure and organization within Figma. Pages are the last level of structure.



Page

- You can use Pages to:
 - Iterate on different versions of a concept or idea
 - Break up designs by platform or device
 - Group designs by feature or product
 - Arrange Components and Styles into categories
 - Track each stage of the design process
 - Let engineers know what's ready for development
 - Present ideas or versions to collaborators
 - Build Atomic Libraries and Design Systems
 - Create different Prototyping flows for the same designs